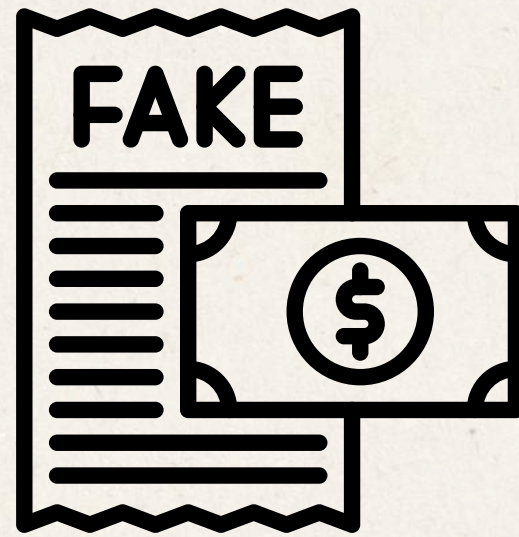




DSA GROUP 3

Credit Card Fraud Detection ML Model

PRESENTED BY:

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Lija
Benedict
Jing En
Ryan

PRESENTED TO:

SIM DAC EXCOs

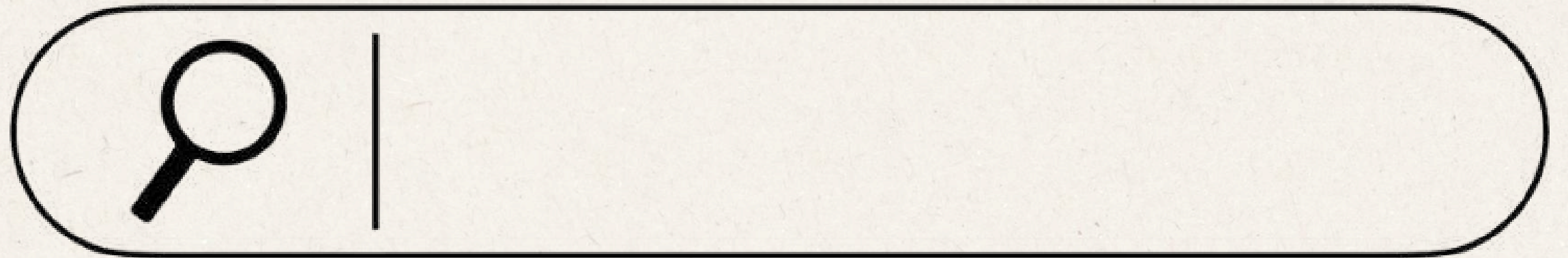


CREDIT CARD FRAUD DETECTION MODEL

A Machine Learning Model That Predicts Credit Card Fraudulent Transactions



INTRODUCTION



Problem Statement

Context:

Financial institutions process large volumes of transactions daily, and even a small number of fraudulent cases can lead to major financial losses.

Goal:

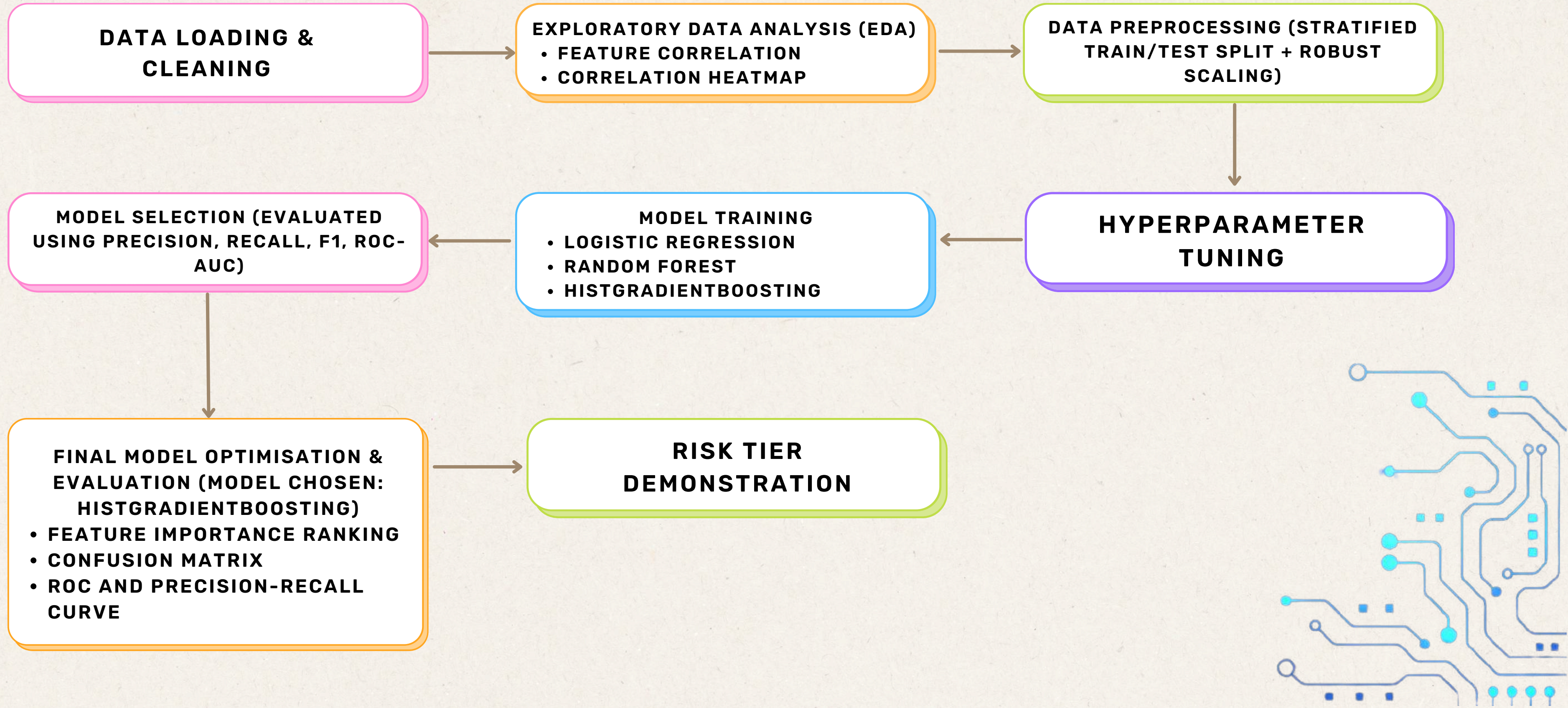
Build a machine learning model that detects fraudulent credit card transactions using anonymised features (V1–V28), transaction amount, and related attributes. The dataset is highly imbalanced, requiring careful preprocessing and evaluation.

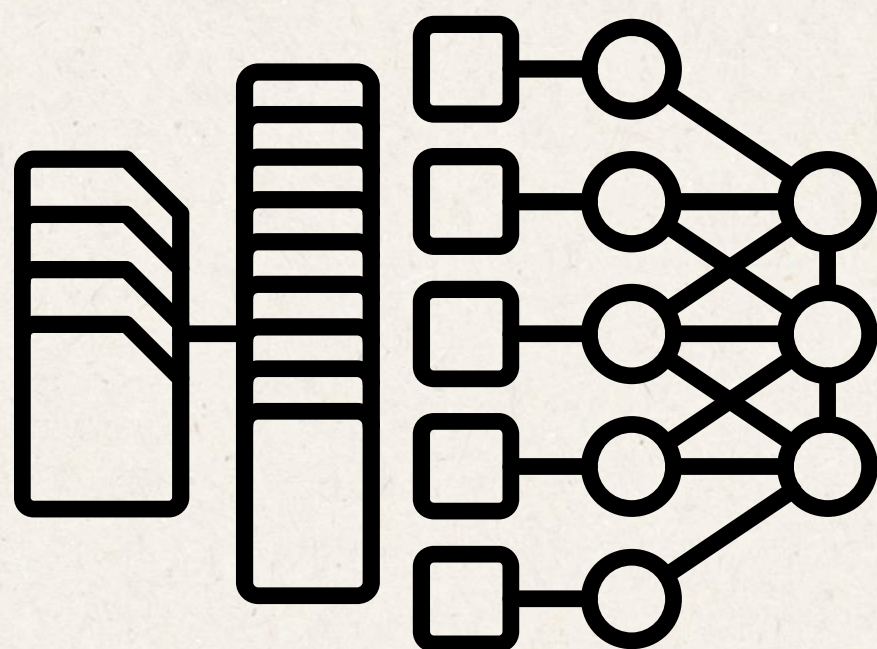
Key Challenge:

Features are anonymized (PCA components) - limits interpretability and prevents domain-specific feature engineering.



Workflow Diagram

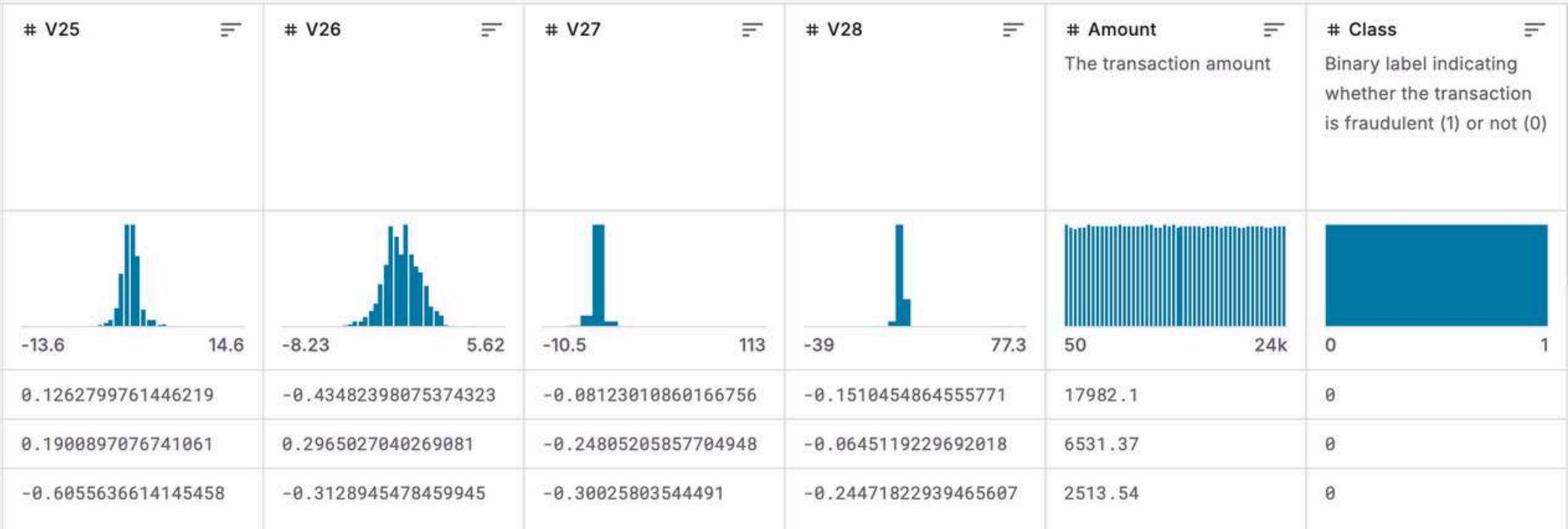
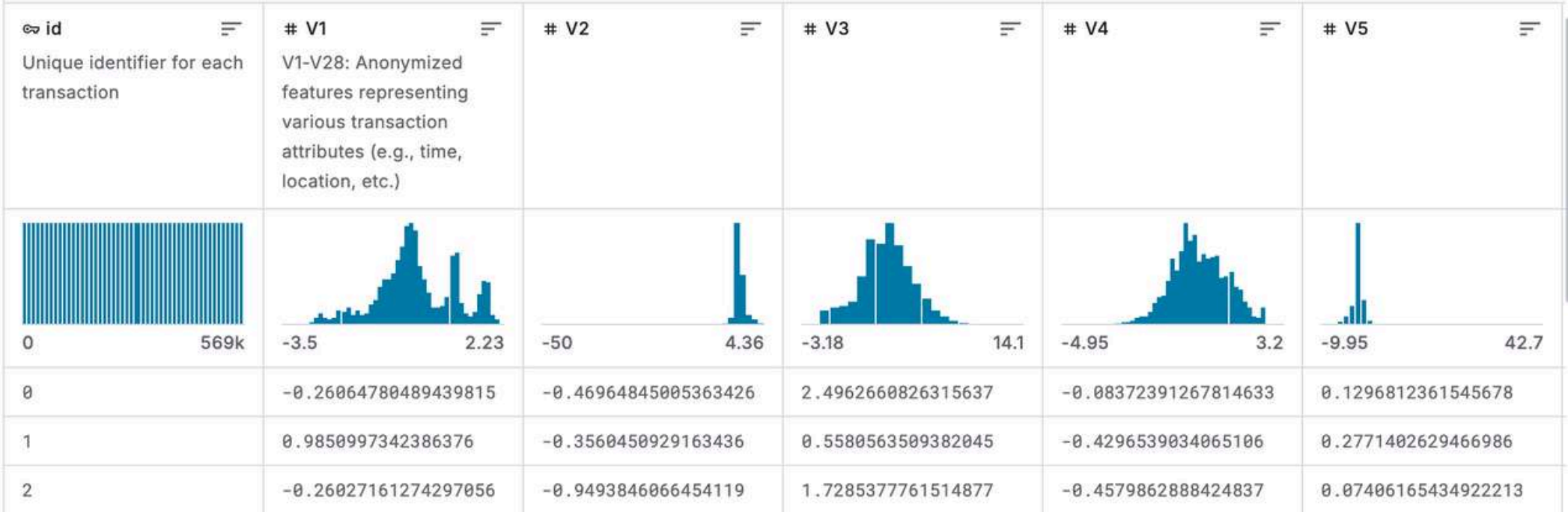




EXPLORATORY DATA ANALYSIS

Visualising the dataset

Dataset Summary & Class Distribution



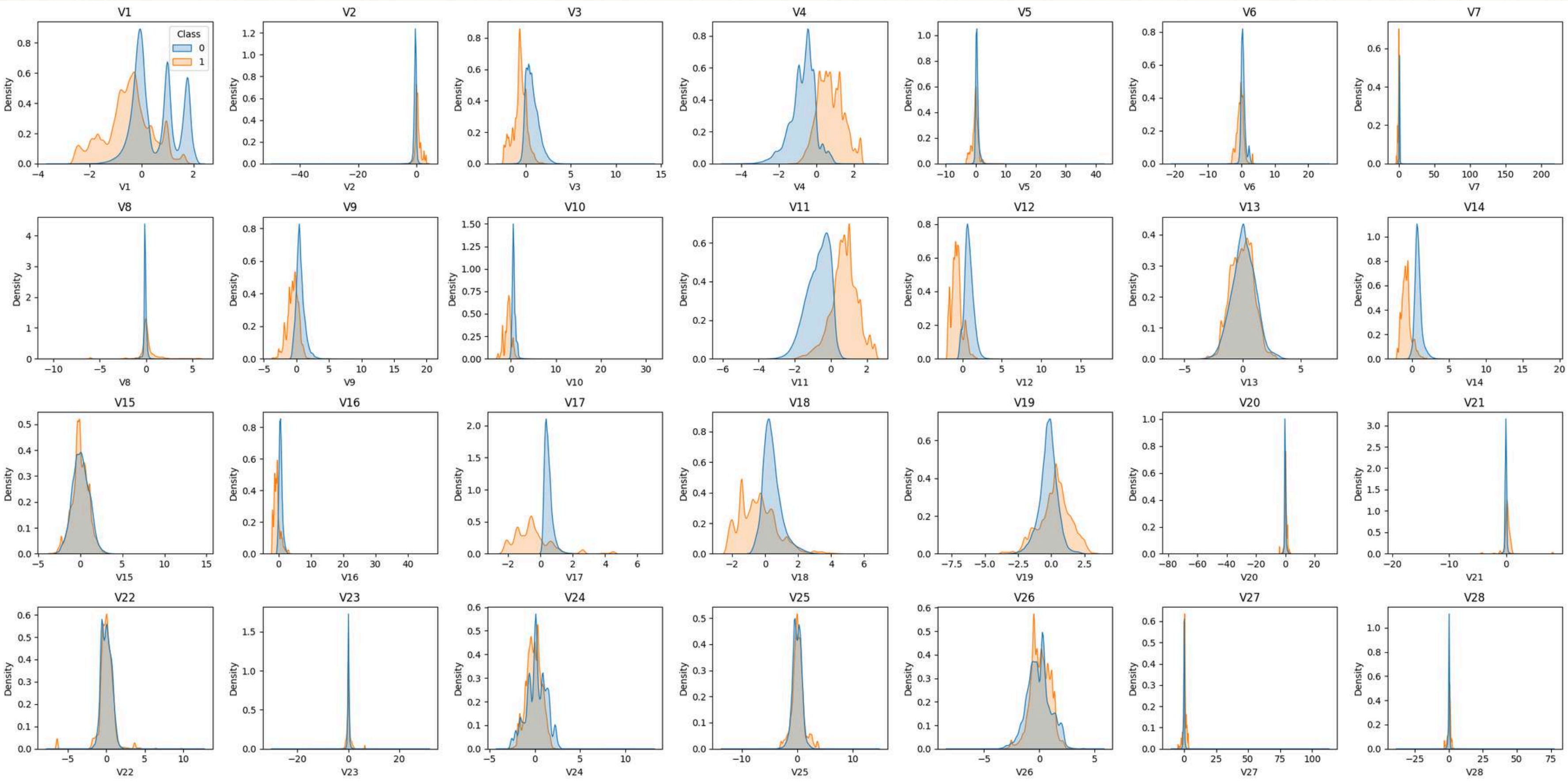
```
Class
0    284315
1    284315
Name: count, dtype: int64
```

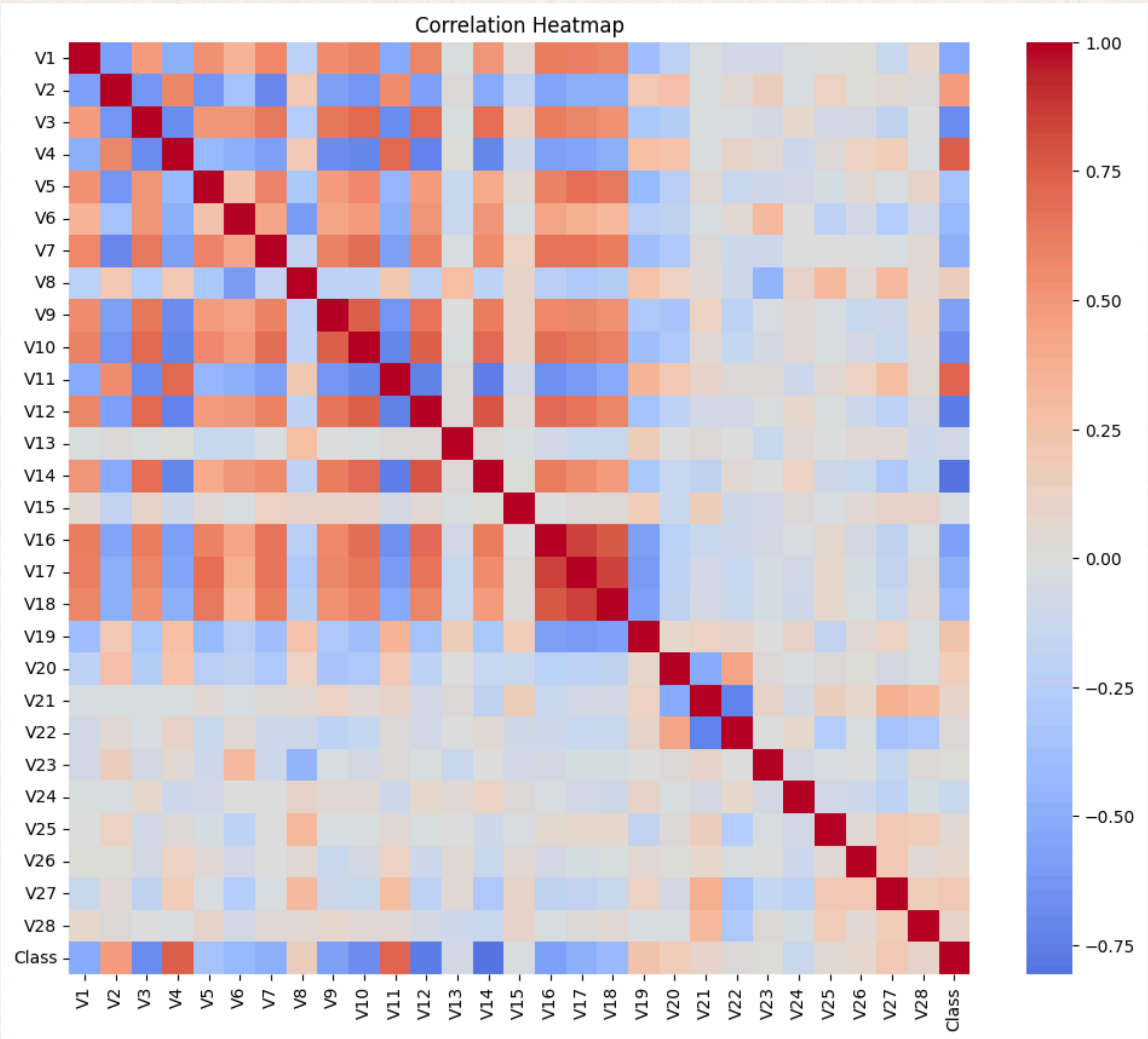
```
0 Missing values
0 Duplicates

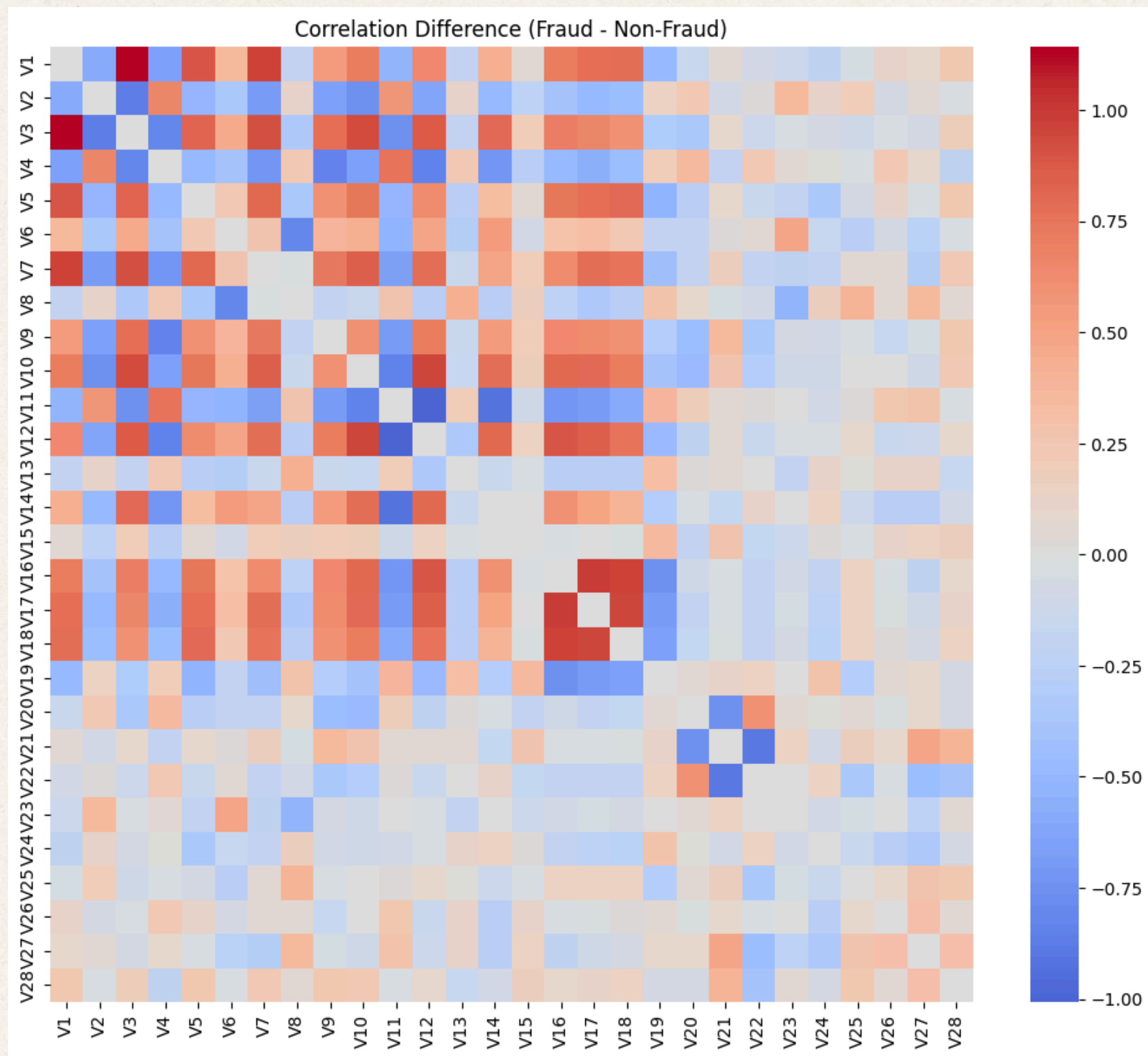
New Shape After Cleaning: (568630, 30)
```

	Class	0	1
V1	count	284315.000000	284315.000000
	mean	0.505761	-0.505761
	std	0.823780	0.899890
	min	-3.495584	-2.637662
	25%	-0.137680	-0.987457
...	...	...	...
Amount	min	50.120000	50.010000
	25%	6034.540000	6074.640000
	50%	11996.900000	12062.450000
	75%	18040.265000	18033.780000
	max	24039.930000	24039.930000

CLASS-WISE DISTRIBUTION OF ANONYMIZED FEATURES (V1-V28)



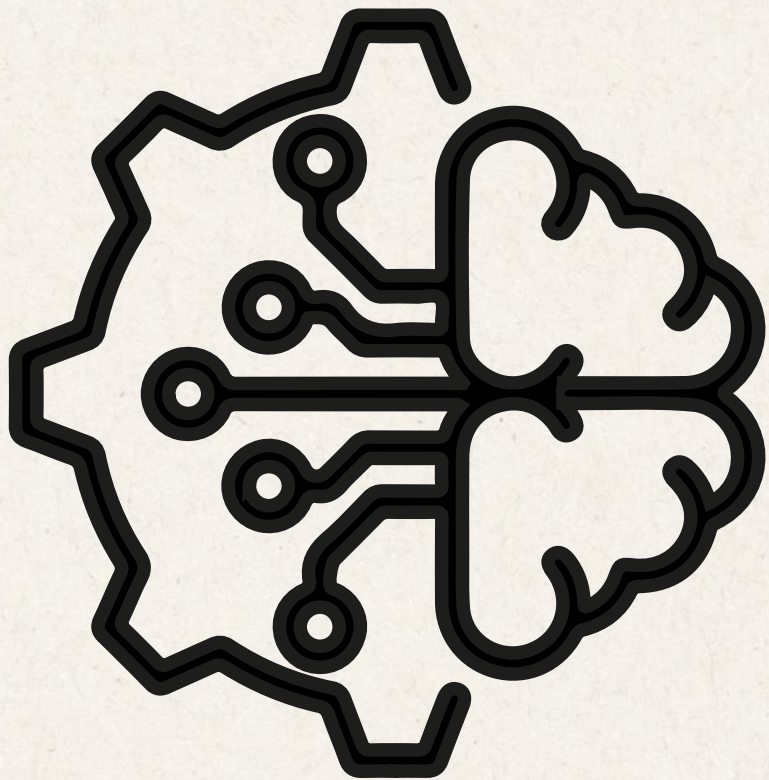




	Feature	Correlation_Difference_Score
0	V3	13.215989
1	V10	13.089119
2	V12	12.983935
3	V7	12.414526
4	V17	12.109052
5	V16	12.003771
6	V1	11.800583
7	V11	11.722775
8	V5	11.627696
9	V9	11.606788
10	V18	11.367889

MODEL SELECTION

supervised learning, binary classification task



Data Preprocessing:

- 1. Removed ID column, Checked missing values & duplicates
- 2. Stratified 80/20 Train-Test Split
- 3. Applied **RobustScaler** to handle outliers
- 4. V-features required scaling due to wide distribution

```
Class Distribution:
  Legitimate (0): 284315 (50.00%)
  Fraud (1): 284315 (50.00%)

Train Shape: 454904 samples
Test Shape: 113726 samples
Train Fraud cases: 227452 (50.00%)
Test Fraud cases: 56863 (50.00%)
```

Models tested before Selection:

- 1. **Logistic Regression**
 - Quick baseline model
 - Lower recall → missed more fraud cases
- 2. **Random Forest**
 - Good (or stronger) performance
 - Slight overfitting on imbalanced data
 - Higher inference cost
- 3. **HistGradientBoostingClassifier**
 - Best ROC-AUC & Recall
 - Handles imbalance + outliers well
 - Captures non-linear relationships
 - Chosen as final model

	Logistic Regression	Random Forest	HistGradient Boosting
Strengths	Fast training, highly inpretable	Handles non-linearity, robust to noise	Captures complex patterns, efficient
Weaknesses	Cannot capture complex non-linear patterns	Higher variance	Less interpretable
Overall Performance	Low performance	Medium performance	Highest recall and F1-score

Metrics Used and Why?

1. ROC-AUC

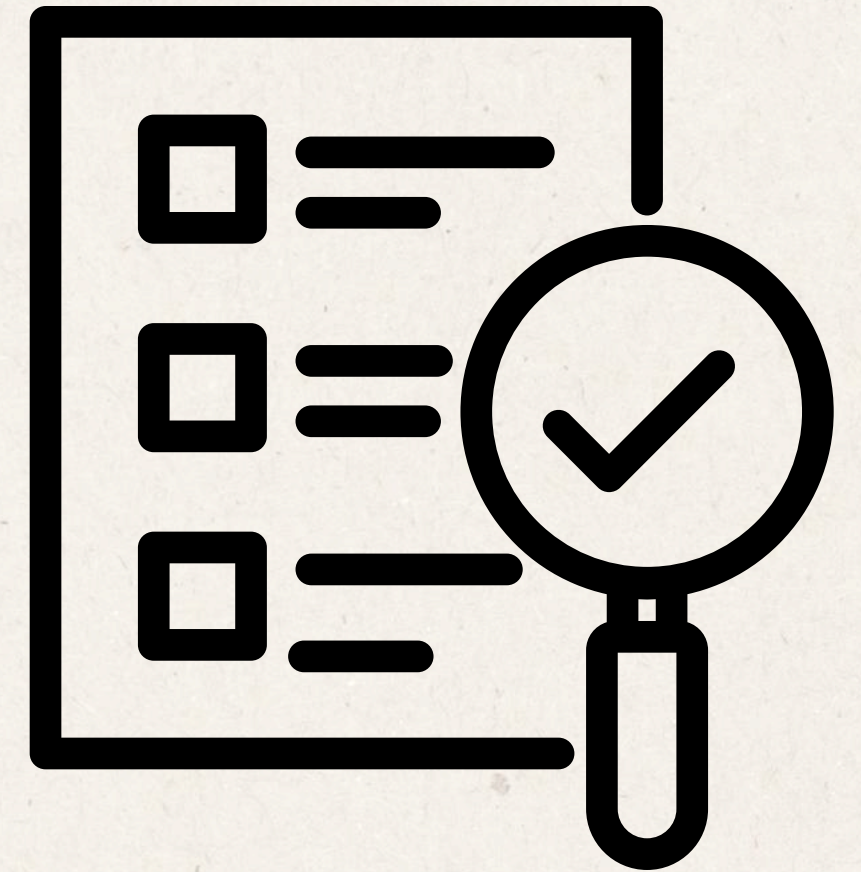
- Main metric for model tuning (via RandomizedSearchCV)
- Measures how well the model separates fraud vs. non-fraud
- Works well for imbalanced datasets

2. Recall

- Measures how many fraud cases were correctly detected
- Critical because missing fraud is more costly than false alarms

3. Precision

- Measures how many predicted frauds were actual fraud
- Helps reduce unnecessary investigations



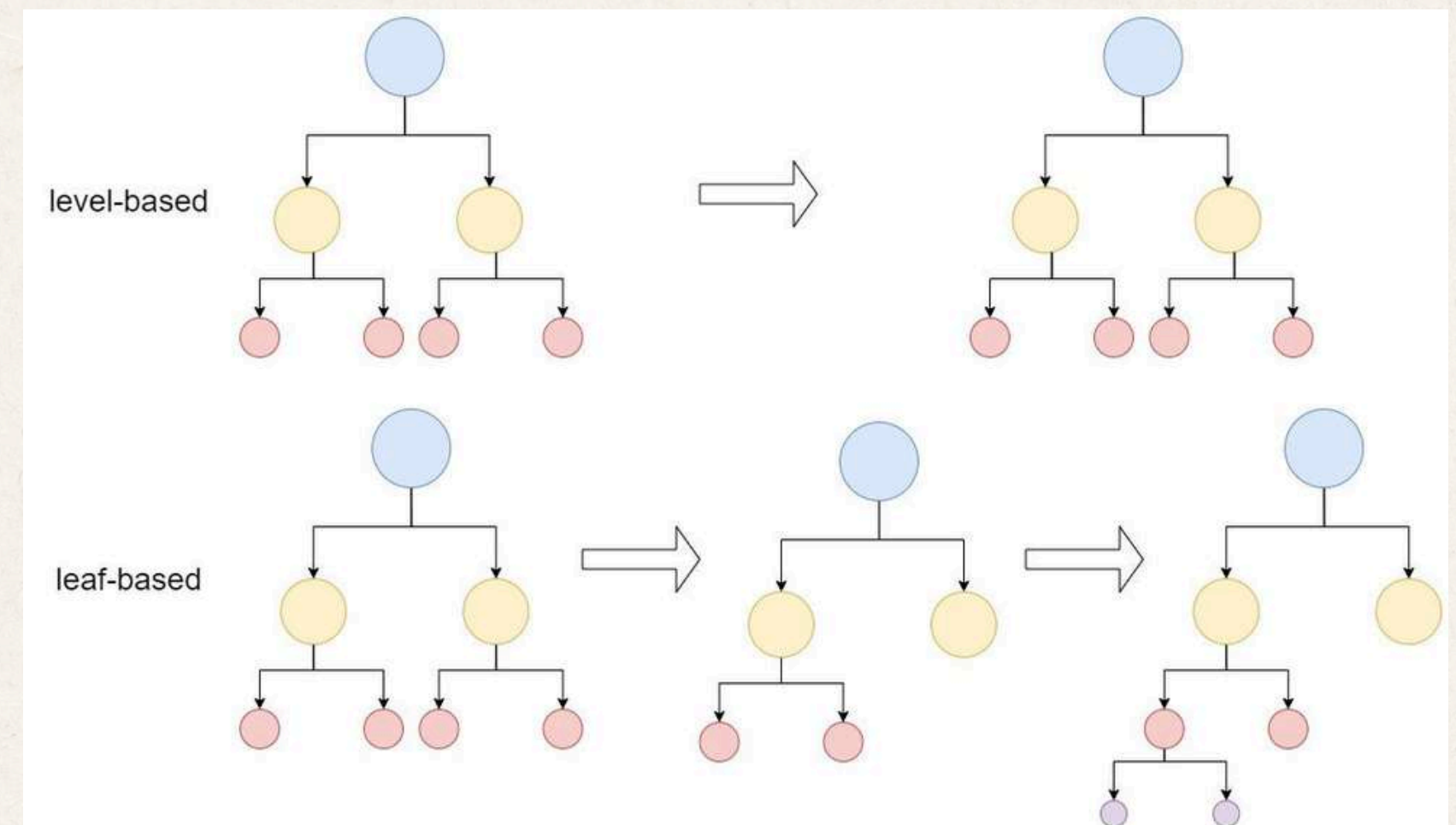
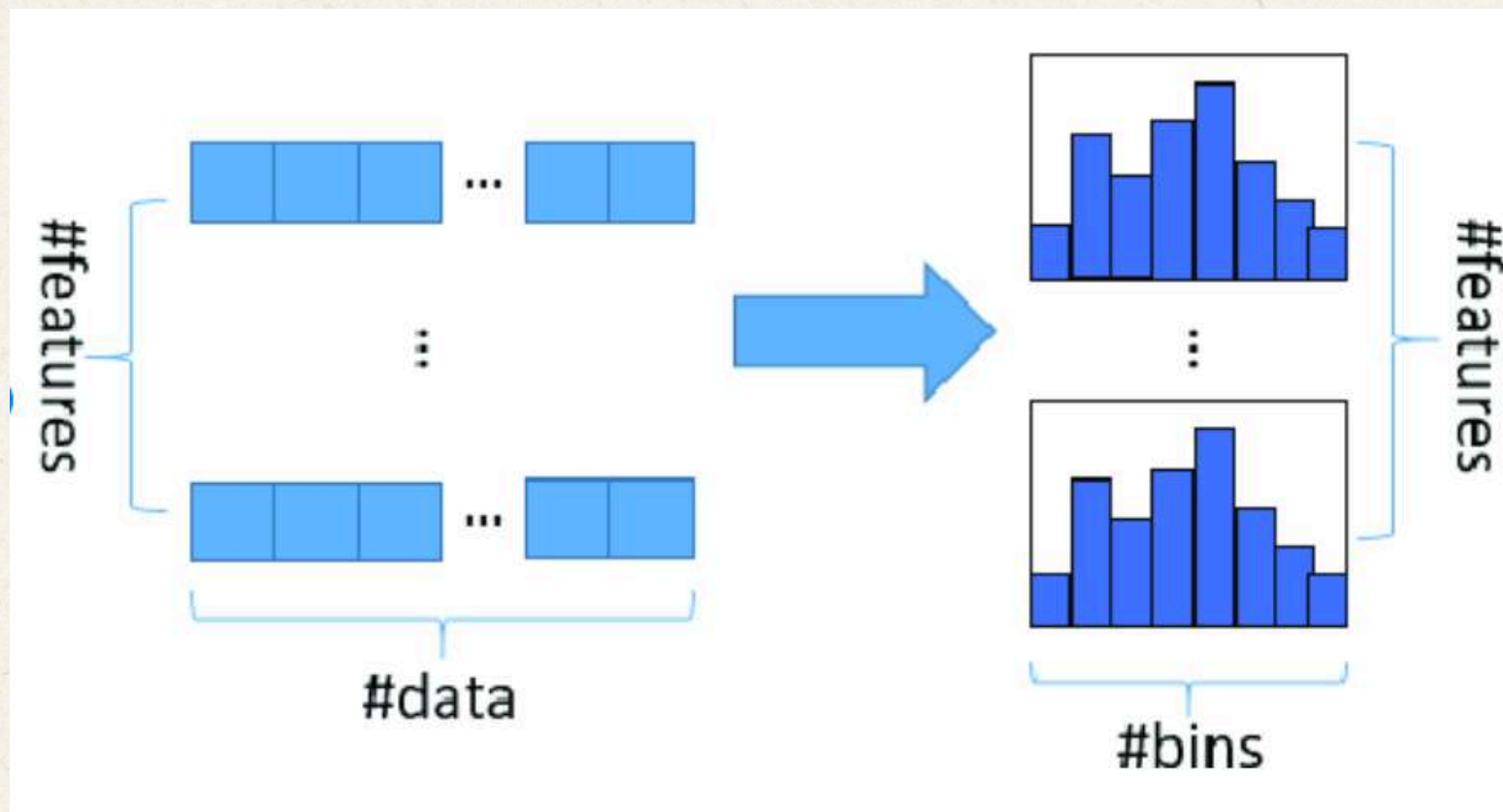
What's HistGradientBoostingClassifier?

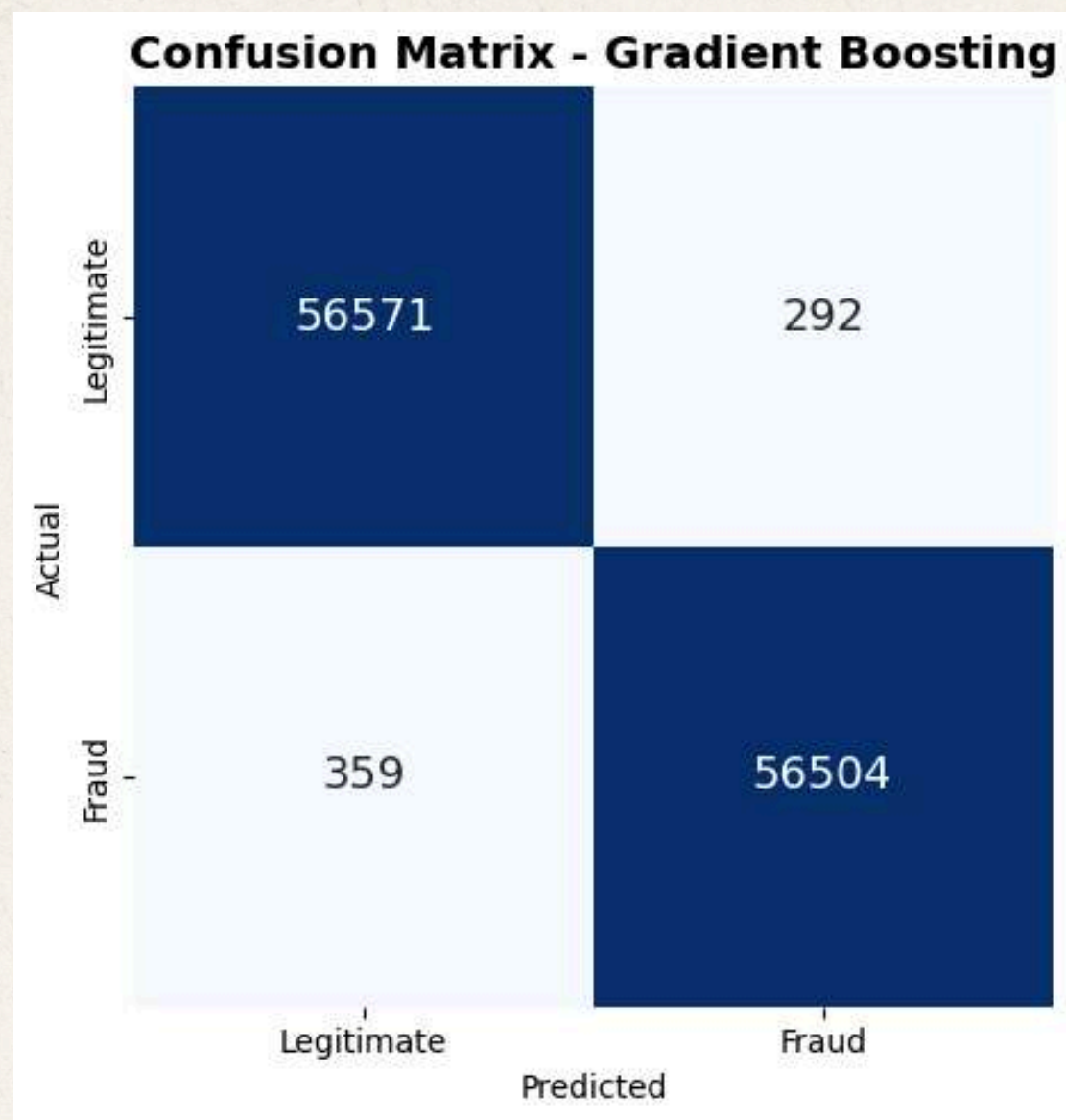
```
from sklearn.ensemble import HistGradientBoostingClassifier
```

Scikit-learn's Histogram-based Gradient Boosting Classification Tree leverages histogram-based techniques to enhance the efficiency and scalability of gradient boosting, this estimator is much **faster** than GradientBoostingClassifier for **big datasets** ($n_samples \geq 10\,000$)

Gradient boosting works by **ensembling** many decision trees in order to perform regression or classification. However, unlike random forest, gradient boosting **grows trees sequentially**, iteratively growing trees based on the residuals of the previous tree. Doing so allows gradient boosting to focus on particularly tricky observations and yields an extraordinarily powerful ensemble of trees.

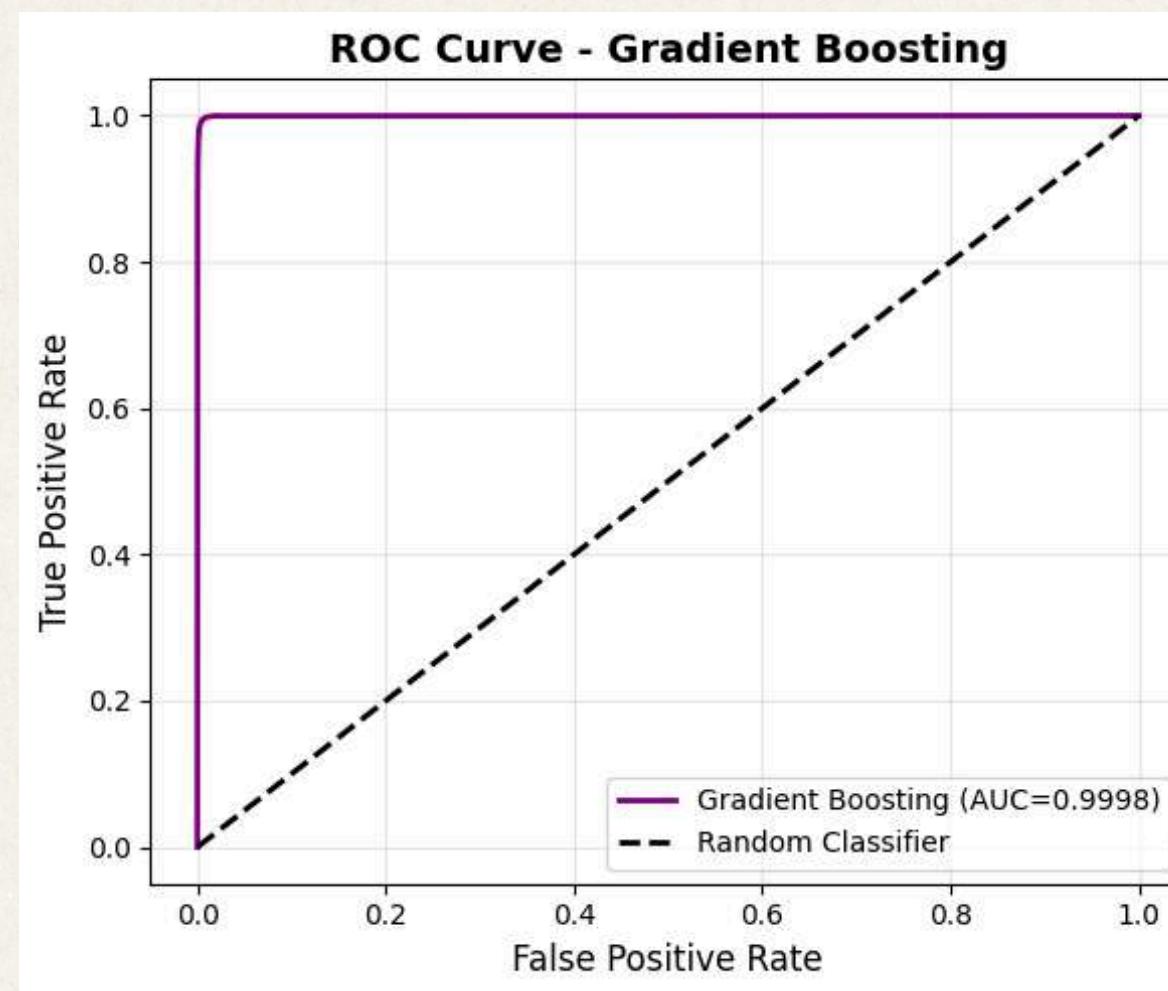
Implementation is inspired by LightGBM.



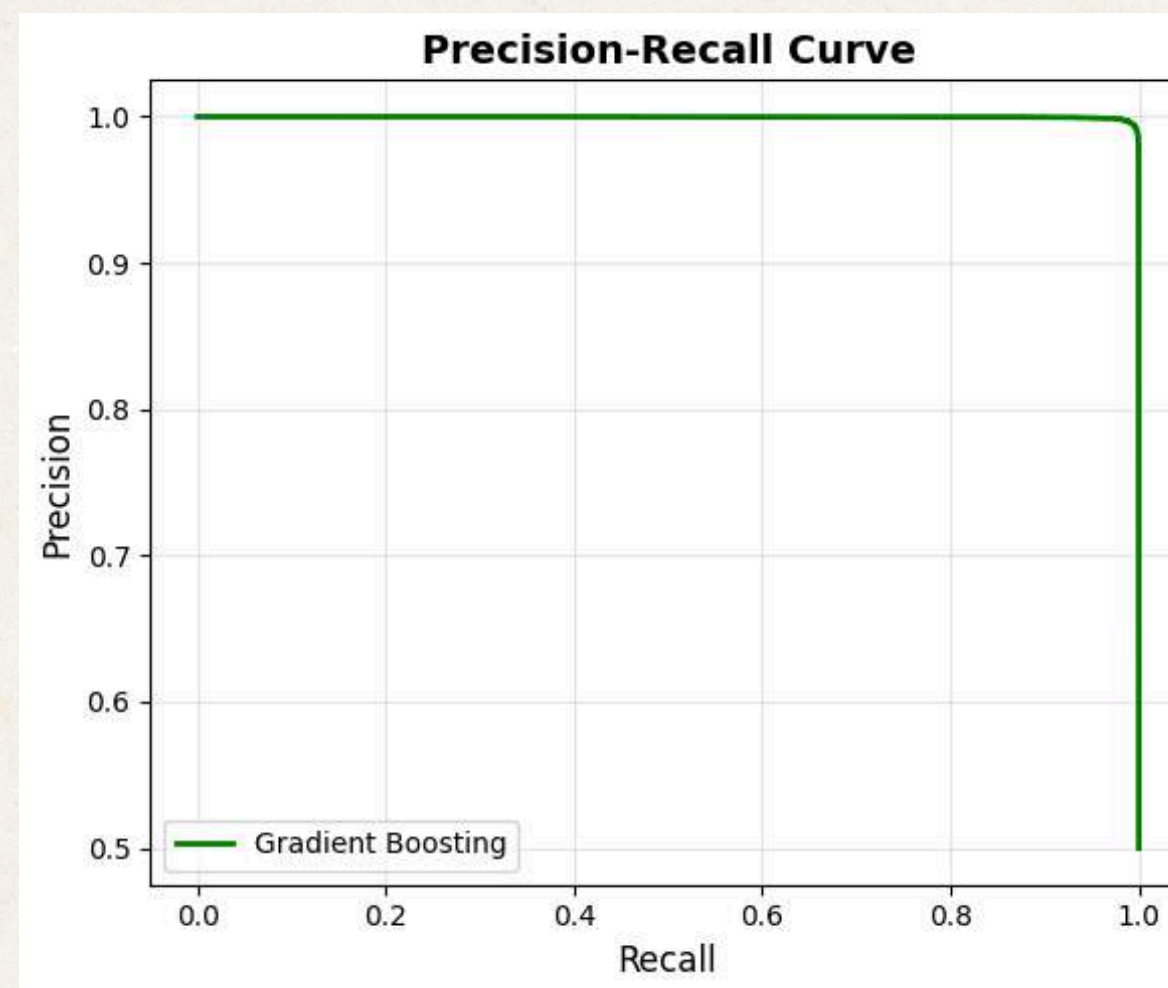


Confusion matrix: How well our model distinguishes between legitimate and fraudulent transactions - model has strong overall detection ability

- Catches >99% of fraudulent transactions.
- Low False Alarms: Only 292 legitimate transactions incorrectly flagged.
- Low Miss Rate: Only 359 frauds missed



ROC Curve (AUC = 0.9998)
Shows separability between fraud and legitimate transactions - model makes correct predictions



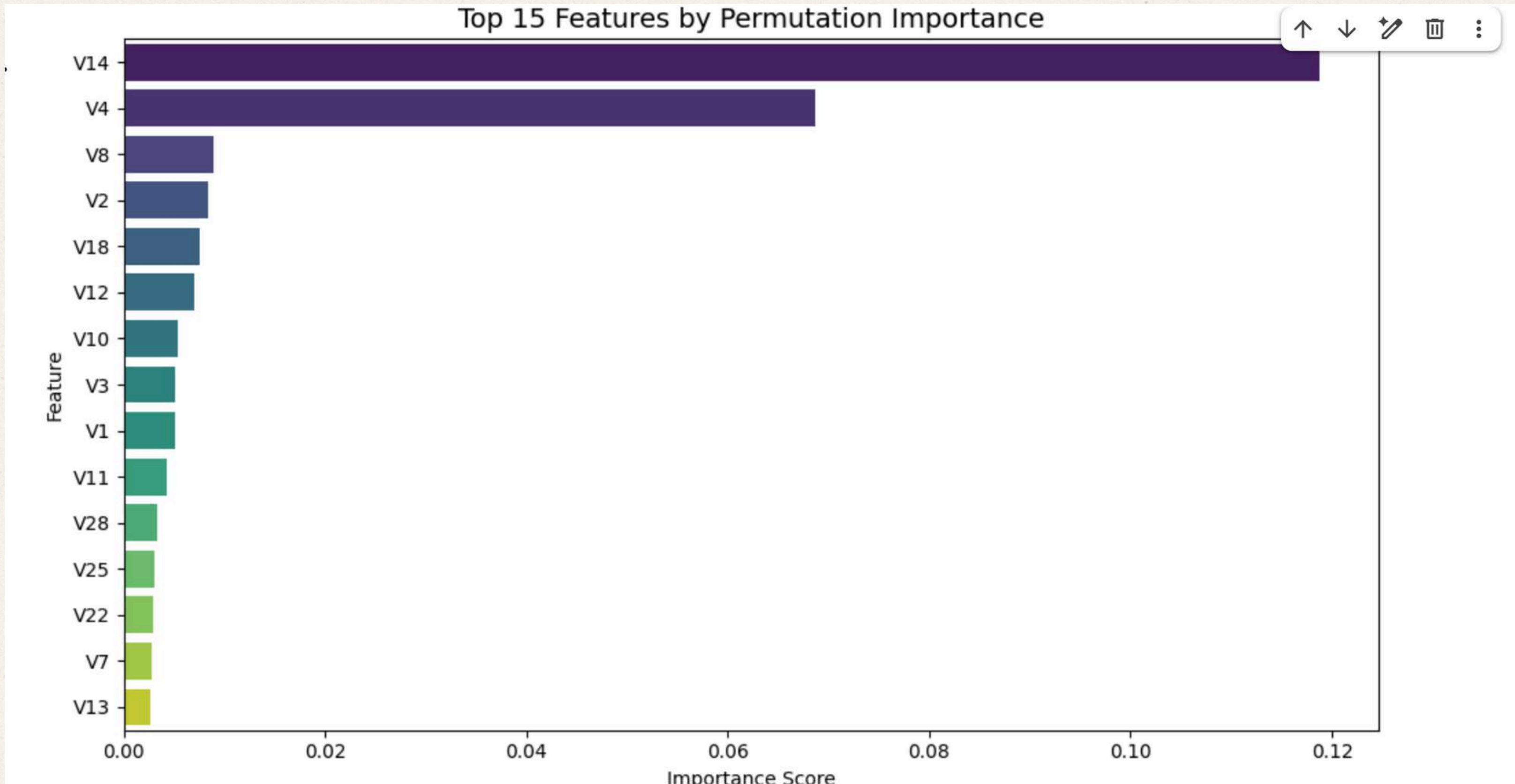
Precision-Recall Curve
Indicates reliable fraud flagging (but will be more useful in a realistic scenario with imbalanced dataset)

Both: shows model is robust across metrics!

Feature Importance — What & Why

- Identifies which features contribute most to fraud predictions
- Helps explain model decisions
- Important for transparency in financial applications
- Use Permutation Importance for HistGradientBoosting

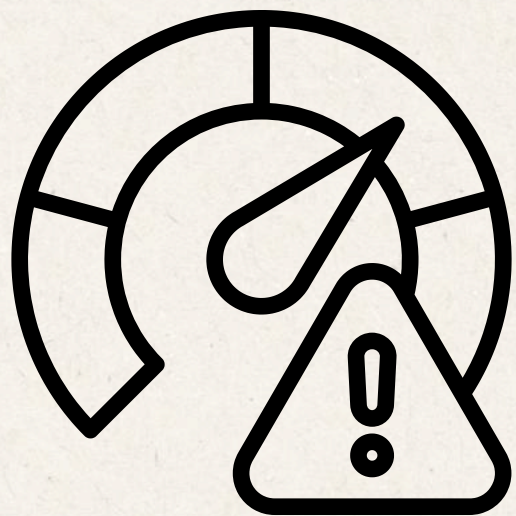
Find The Most Important Features





THE RISK ENGINE

A Concept That Uses Probability To Categorise Risk Level



Fraud Vs Not Fraud

Problem with Binary Prediction (Yes Or No)

- Forces the Model to make a hard decision
- Strict Thresholds: Prone to committing False Positives
- Loose Thresholds: Prone to committing False Negatives

For Example:

Model Detects 49% Risk = Not a fraudulent transaction

Model Detects 51% Risk = Fraudulent transaction

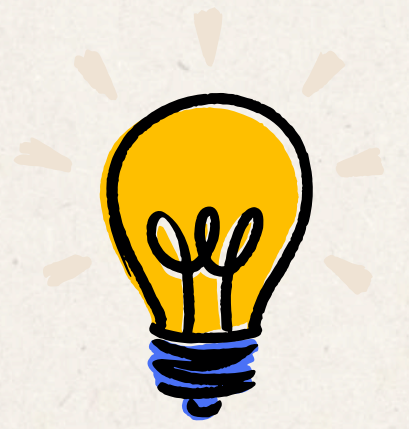
Fraudulent Transactions cannot be predicted as Fraud or Not Fraud at least accurately.

Our finding above show:

It is not practical to use Binary Prediction in real world context



THE SOLUTION



Give the model a new transaction

Model calculates the probability of it being fraud

Model categorises the transaction into the Risk Tier Scoring System

RISK TIER SCORING SYSTEM

Low Risk: < 10% Probability

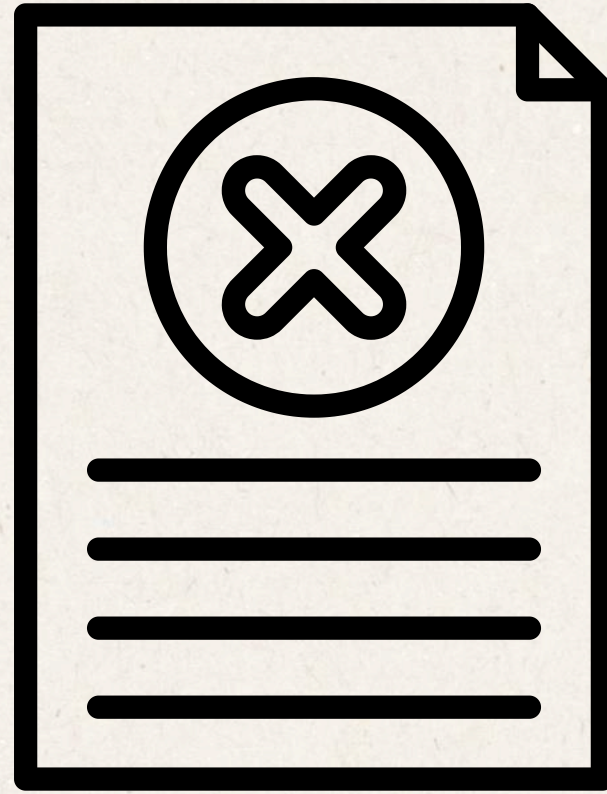
Medium Risk: < 50% Probability

High Risk: < 90% Probability

Critical Risk: $\geq$ 90% Probability

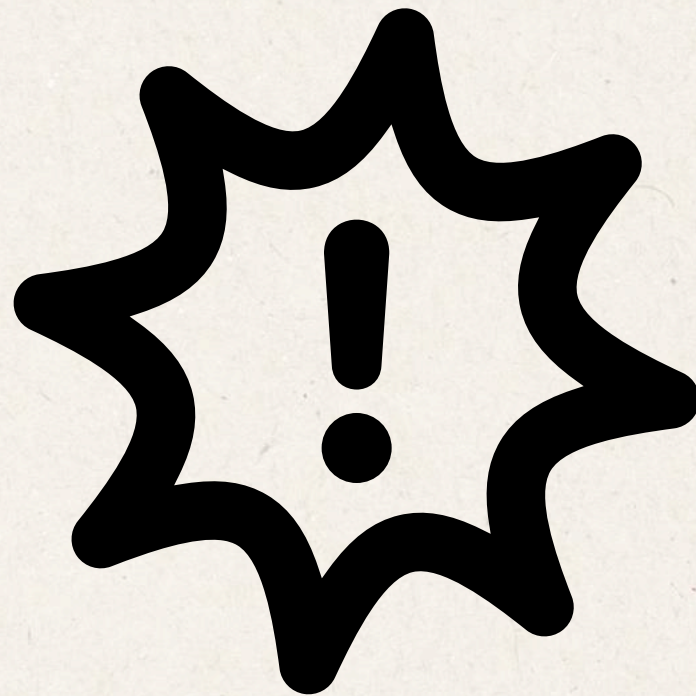
● FRAUD DETECTION REPORT

ACTUAL STATUS:	LEGITIMATE (True Label: 0)
MODEL PREDICTION:	LEGITIMATE
FRAUD PROBABILITY:	2.33%
RISK TIER:	● Low Risk



LIMITATIONS

The Barriers To The Model



The “Black Box” Problem

- Data uses anonymised PCA features (eg. V1-V28)
- Unable to fully interpret why a transaction is fraud in real world context
- Dataset is in coefficients and have no real value

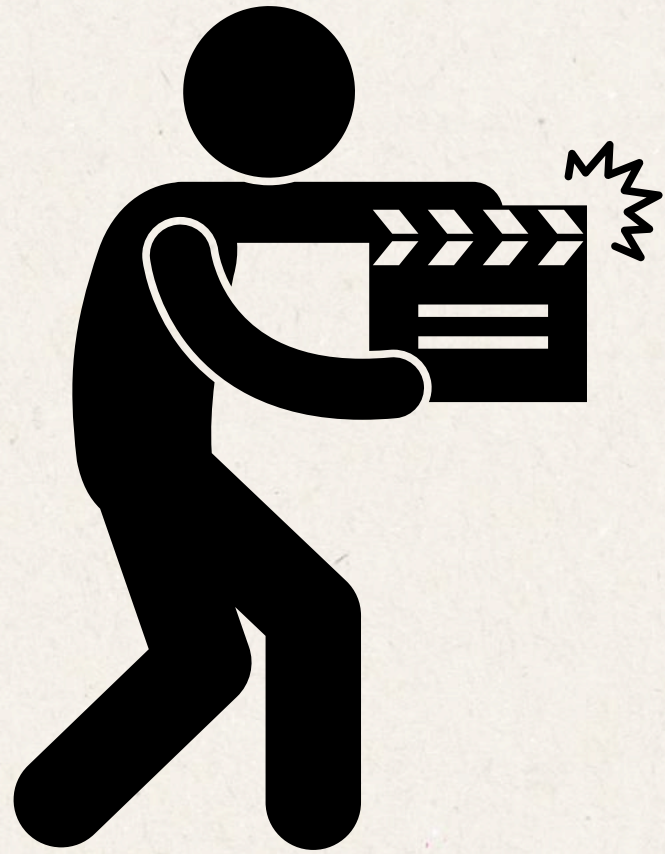
Aging Model

- Model is trained only on a specific dataset - *creditcard2023_csv*
- Model may be obsolete as fraud tactics evolve

Individual Behaviour Patterns

- Dataset used to train model does not account for individual spending habits
- Model might identify non-fraud transaction as high-risk





CONCLUSION

Wrapping the presentation up



Key Achievements



High-Performance Detection

- HistGradientBoosting trained model was capable of handling complex, high volume data
- Achieved decent statistics in terms of accuracy (eg. F1 score etc)

Beyond Binary

- Solved the problem of hard model being forced a “yes and no” answer
- Opened more options for call-to-action (eg. If classified as critical risk, transaction to be blocked)

Decoded Anonymised Patterns

- Faced a major challenge with the lack of raw business context
- Proved that mathematically transformed data can still be precise

Future Scope:

- Implement an automated retraining pipeline to adapt to new fraud tactics
- Risk Tier System to have a call-to-action (eg. Moderate Risk to have notifications sent to user)
- Investigate de-anonymisation techniques to map mathematical features



QNA

Any Questions?

